Qiyang Hong

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EDUCATION

- **Ph.D. in Biomedical Engineering (Medical AI)**, Tsinghua University & Peking Union Medical College, Beijing, China (Sept 2023–June 2026)

  Advisor: Prof. Erping Long (Researcher; National Overseas Young Talent)

  Core Expertise: Foundation models, LLMs, multimodal learning, temporal modeling; deep phenotyping, clinical AI; model interpretability; high-performance computing (4×A6000, 2×H100)

- **M.S. in Biochemistry and Molecular Biology**, Xiamen University, Xiamen, China (Sept 2014–July 2017)

  Advisor: Prof. Ning-Shao Xia (Academician, Chinese Academy of Engineering)

  Specialization: Transcriptomics, viral immunology, structural biology, computational molecular analysis

- **B.Eng. in Bioengineering**, Jimei University, Xiamen, China (Sept 2010–July 2014)
- **B.S. in Computer Science and Technology,** The Open University of China, Beijing, China (Sept 2023–2026)

PUBLICATIONS

*Equal contributor; #Corresponding author. Journal Impact Factors (JIFs) are 2025 JCR values.

Published Articles

2026

1. Yi E*, Cui J*, Wang H*, **Hong Q***, ..., Long E#, Ran P*. Multiomics Mendelian randomization identifies serpin family G member 1 as a chronic obstructive pulmonary disease modulator. *Signal Transduction and Targeted Therapy.* 2026;11:34. doi:10.1038/s41392-025-02547-7. JIF (2025): **81.2**.
2. **Hong Q***, Wang C*, ..., Long E#. A foundational model encodes deep phenotyping data and enables diverse downstream applications. *npj Digital Medicine*. 2026;9 (online ahead of print). doi:10.1038/s41746-026-02736-w. JIF (2025): **18.0**.
3. Sun X*, **Hong Q***, ..., Wan P#, Long E#. Model confrontation and collaboration: a debate intelligence framework for enhancing medical reasoning in large language models. *Cell Reports Medicine*. 2026;7:102547. doi:10.1016/j.xcrm.2025.102547. JIF (2025): **14.0**.
4. Wu W*, Zhou M*, Chen J, Wang J, Hong Q, ..., Pang J#, Long E#, Wang J#. Deep multi-omics profiling reveals three molecular subtypes of chronic obstructive pulmonary disease in a unique biomass-exposed Chinese population. *Med*. 2026;7:101033. doi:10.1016/j.medj.2026.101033. JIF (2025): **13.3**.
5. Pang J*, Wang J*, Wu S, Hong Q, ..., Long E#. GWAS meta-analyses with a new Chinese population expand the genetic architecture and improve ancestry-specific genetic risk prediction of chronic obstructive pulmonary disease. *Genomics, Proteomics & Bioinformatics*. 2026 (accepted in principle). JIF (2025): **7.2**.
6. Pang J*, Ren X*, **Hong Q***, Huang K, Wang J, Li D, Huang X, ..., Liang C#, Long E#, Wang J#, Yang T#, Wang C. Proteo-metabolomic insights into the progression of chronic obstructive pulmonary disease and lung function decline. *Respiratory Research*. 2026 (Accepted). JIF (2025): **5.7**.

2025

7. Long Y*, Ni X#*, Chen T*, Huang X, Hong Q, ..., Zhang T#, Long E#. Genetic determinants of gene expression noise and its role in complex trait variation. *Cell Reports*. 2025;44:116612. doi:10.1016/j.celrep.2025.116612. JIF (2025): **7.7**.

2024

8. Lin K*, **Hong Q***, ..., Chen M#. Cervical HPV infection and related diseases among 149,559 women in Fujian: an epidemiological study from 2018 to 2023. *Frontiers in Microbiology.* 2024;15:1418218. doi:10.3389/fmicb.2024.1418218. JIF (2025): **5.8**.

2022

9. Huang Y[*], Li Z[*], Hong Q, ..., Yu H[#], Li S[#], Xia N[#]. A stepwise docking molecular dynamics approach for simulating antibody recognition with substantial conformational changes. *Computational and Structural Biotechnology Journal*. 2022;20:710–720. doi:10.1016/j.csbj.2022.01.012. JIF (2025): **4.8**.

2017

10. Xu L[*], Zheng Q, Li S, He M, Wu Y, Li Y, Zhu R, Yu H, Hong Q, ..., Xia N[#]. Atomic structures of Coxsackievirus A6 and its complex with a neutralizing antibody. *Nature Communications*. 2017;8:505. doi:10.1038/s41467-017-00477-9. JIF (2025): **18.1**.

2016

11. Li Z[*], Yan X[*], Yu H[*], Wang D, Song S, Li Y, He M, Hong Q, ..., Baker T[#], Li S[#], Xia N[#]. The C-terminal arm of the human papillomavirus major capsid protein is immunogenic and involved in virus–host interaction. *Structure*. 2016;24:874–885. doi:10.1016/j.str.2016.04.008. JIF (2025): **4.6**.

Manuscripts Under Review / Submitted

1. **Hong Q**[*], Xu S[*], Zhang Q[*], ..., Long E[#]. Educator–LLM collaboration strategies for discharge education in hematopoietic stem cell transplantation: a randomized controlled trial of real-time prompting. *NEJM AI* (Under Review).
2. Ni X[*#], Chen T[*], Hai E[*], Huang Z, Li Y, Sun X, Hong Q, Long E[#]. A mediation-informed causal pathway powers a large language model for individualized health advisory. *Nature Aging* (Under Review).
3. Wu F[*], **Hong Q**[*], Pang J[*], ..., Long E[#], Ran P[#], Yi E[#]. Cough, sputum, and CD164: novel risk factors and biomarkers for COPD and lung function decline. *Respiratory Research* (Submitted).

TECHNICAL SKILLS

- **Artificial Intelligence & Machine Learning:**
  Neural networks, Transformer architectures, foundation models, LLMs, multimodal learning, temporal modeling, representation learning, model interpretability.
- **Model Training & High-Performance Computing:**
  PyTorch, CUDA, distributed training, GPU cluster management (Linux HPC, A6000 nodes), job scheduling, system optimization, large-scale data preprocessing, performance profiling.
- **Bioinformatics & Genomic Analysis:**
  WES/WGS pipelines, RNA-seq, SNP/Indel/CNV detection, variant annotation and ACMG classification, immunogenomics, neoantigen prediction, clinical diagnostics workflow development.
- **Programming & Data Engineering:**
  Python, NumPy, pandas, scikit-learn; workflow automation; reproducible pipelines; Django-based web applications.
- **Data Visualization & Network Analysis:**
  matplotlib, seaborn, igraph, networkx; interactive visualization for clinical and genomic datasets.

CERTIFICATIONS & LICENSES

- **Health Professional Qualification (Clinical Laboratory / Medical Testing)**
  National Health Authority of China — 2012
  (National-level professional technical qualification)
- **Clinical PCR Laboratory Technician Certification**
  Fujian Provincial Clinical Laboratory Center — Sept 2020
  (Authorized training for clinical gene amplification laboratories)
- **Bioinformatics Engineer Certification**
  ICT Support / ICTTT — Jan 2015

PROFESSIONAL EXPERIENCE

- **Group Leader, Molecular Diagnostics**
  Affiliated Hospital of Putian University, Prenatal Diagnosis Center
  Aug 2020 – Jul 2023

- **Senior Bioinformatics Engineer (Part-time)**

  Jinfeng Biotechnology Co., Ltd., Data & Information Department

  Nov 2020 – Jan 2022

- **Bioinformatics Supervisor**

  Berry Genomics Co., Ltd.

  Jul 2017 – Aug 2020

PROFESSIONAL ACTIVITIES

Co-reviewer (with Prof. Erping Long) : Nature Medicine, Nature Biomedical Engineering, Frontiers in Aging Neuroscience

RESEARCH PROJECTS & TECHNOLOGY DEVELOPMENT

- **Temporal Foundation Models for COPD Progression**

  National Science and Technology Major Project (Youth Scientist Program), 2025–2028

  Developing longitudinal multi-omics integration and temporal neural networks to decode COPD progression, identify molecular signatures, and support early risk prediction.

- **Biomedical Foundation Model (UK Biobank)**

  Built a Transformer-based foundation model trained on deep phenotyping data from >500,000 participants, enabling disease prediction, multimorbidity analysis, and patient stratification across 289 conditions.

- **Automated Clinical WES Analysis Pipeline**

  Developed an end-to-end workflow from FASTQ to SNP/Indel/CNV detection, annotation, ACMG classification, and automated reporting; deployed in clinical settings with version-controlled pipelines.

- **Neoantigen Prediction and Immunogenomics Pipeline**

  Integrated WES and RNA-seq pipelines with NetMHC/MHCflurry to identify candidate neoantigens for immunotherapy and tumor profiling.

- **WES Clinical Interpretation and Visualization Platform**

  Implemented a Django-based interactive web platform enabling clinicians to visualize variants, coverage, CNVs, and Sanger traces, supporting clinical genetic diagnosis.

- **Consumer Genomics Analysis Framework**

  Contributed to the development and maintenance of large-scale production pipelines for SNP genotyping and ancestry/trait reporting in collaboration with consumer genomics companies.

- **Inherited Disease and Metabolic Panel Pipelines**

  Designed and optimized pipelines for thalassemia, SMA, mitochondrial disorders, and mtDNA variant detection to support clinical molecular diagnostics.

- **Custom WES Probe Design Optimization**

  Improved exome capture efficiency by designing probe sets enriched for pathogenic ClinVar regions, enhancing diagnostic yield.

- **High-throughput Transcriptomics Analysis**

  Performed differential expression and KEGG/GO enrichment analysis for bacterial and human datasets under diverse physiological conditions.